MARYAM MIRADI

10X Your AI Solutions for Real-World

(Data Science Roadmap to Excellence)

100 Python Libraries

for Impactful Machine Learning and Deep Learning

by Maryam Miradi

Contents

Thank you for downloading this Guide!
If You have any these questions then this Guide is for you.

1. Misaligned Business Objectives

- "How does this model or solution directly address our core business objectives?"
- "Have we clearly defined the business problem we're trying to solve?"

2. Inefficient Data Handling

- "Have we thoroughly cleaned and preprocessed our data to ensure it's of high quality?"
- "Are we aware of any biases or anomalies in our data that could impact model performance?"

3. Poor Model Performance

- "Have we explored various models and performed hyperparameter tuning to ensure optimal performance?"
- "Do we have a benchmark or baseline model performance for comparison?"

4. Lack of Reproducibility

- "Can someone else easily reproduce our results based on the documentation and code we've provided?"
- "Have we documented all the steps, from data preprocessing to model selection and tuning?"

5. Failure to Detect and Mitigate Bias

- "Have we assessed our model for potential biases against certain groups or individuals?"
- "What steps have we taken to mitigate any identified biases in our model?"

6. Inadequate Evaluation

- "Are we evaluating our model on a diverse set of metrics to understand its strengths and weaknesses fully?"
- "Have we used techniques like cross-validation to ensure our model generalizes well to unseen data?"

7. Deployment Challenges

- "Is our model designed with deployment in mind, considering integration and operational needs?"
- "Have we planned for model monitoring and maintenance post-deployment?"

8. Reduced Trust and Credibility

- "Can we explain how our model makes decisions in a way that's understandable to non-technical stakeholders?"
- "Have we involved stakeholders in the evaluation process to build trust in our model's predictions?"

9. Missed Opportunities for Improvement

- "Are we continuously monitoring model performance and seeking opportunities for improvement?"
- "How responsive are we to new data or changing business needs?"

10. Legal and Ethical Implications

- "Have we considered the legal and ethical implications of our data sources and modeling techniques?"
- "Are we compliant with relevant privacy laws and ethical guidelines in our data science practices?"

My Website

My Linkedin

My Youtube

It's important to realize that progress is Not Linear in a Data Science project.

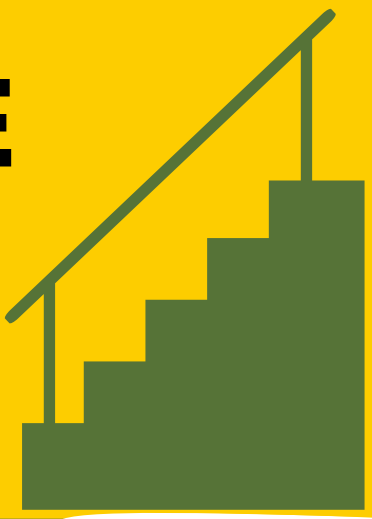
Data Science projects are cyclical and experimental in nature.

As your understanding of the data and problem increase, you'll often revisit earlier approaches and make adjustments.

You may even modify the way you're asking the question to make your outcome more actionable.

Because of this, it's possible to jump back and forth between virtually any of the steps.

DATA SCIENCE
Steps for
IMPACT



BUSINESS UNDERSTANDING

DATA UNDERSTANDING & EDA

DATA PREPARATION

FEATURE ENGINEERING

MODELING

MODEL FINE-TUNNING

EVALUATION & COMPARISON

DEPLOYMENT

EXPLAIN-ABILITY

FAIRNESS

GOAL REACHED

MARYAM MIRADI

1. Business Understanding

While many teams hurry through this phase, establishing a strong business understanding is like building the foundation of a house – absolutely essential.

Steps

1.1 Determine Business Objectives

1.2 Assess Situation

1.3 Determine Data Science Goals

1.4 Produce Project Plan

2. Data Understanding & EDA

Steps

2.1 Collect Initial Data

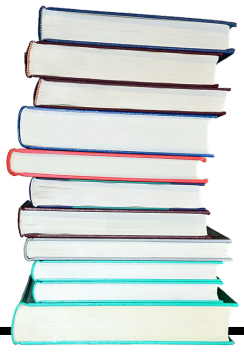
2.2 Describe Data

2.3 Explore Data

2.4 Verify Data Quality

Python Libraries

Plotly
D-Tale
ydata-profiling
Missingno
Dataprep
Lux
Holoviews
PandasGU
NumPy
Pandas
Matplotlib
Seaborn



3. Data Preparation

Steps

3.1 Select Data

3.2 Clean Data

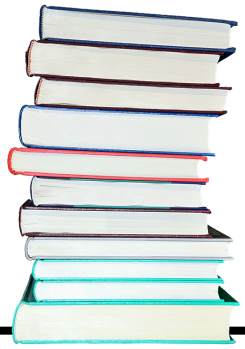
3.3 Construct Data

3.4 Integrate Data

3.5 Format Data

Python Libraries

Feature-engine
Fancyimpute
Missingno
Category Encoders
Imbalanced-learn
PyOD
Great Expectations
Pandas-profiling
Optimus
DataCleaner



4. Feature Engineering

Enhance model performance through the creation, selection, and transformation of features.

Steps

4.1 Feature Creation

4.2 Feature Transformation

4.3 Feature Selection

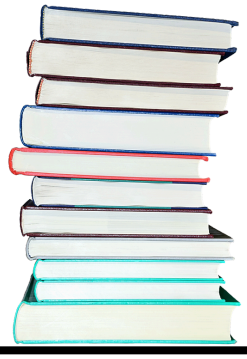
4.4 Feature Encoding

4.5 Feature Scaling

4.6 Feature Evaluation

Python Libraries

Featuretools
TSFresh (Time Series Fresh)
Polars
Category Encoders
Deep Feature Synthesis (DFS) in Featuretools
PyCaret
Boruta
sklearn-pandas
Optimus
AutoFeat



5. Modelling

Steps

5.1 Select Modeling Technique

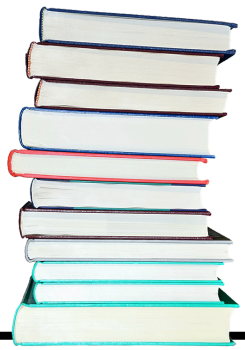
5.2 Generate Test Design

5.3 Build Model

5.4 Assess Model

Python Libraries

Scikit-learn
TensorFlow
PyTorch
XGBoost
LightGBM
spaCy
CatBoost
H2O.ai
Keras
Fast.ai
Transformers (by Hugging Face)



6. Model Fine-Tunning

Steps

6.1 Preparation

6.2 Hyperparameter Space Definition

6.3 Tuning Strategy Selection

6.4 Experimentation Setup

6.5 Execution

6.6 Evaluation and Selection

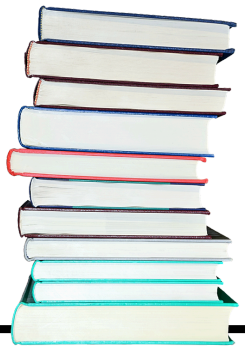
6.7 Refinement (Optional)

6.8 Finalization

6.9 Deployment Preparation

Python Libraries

Auto-Sklearn
TPOT (Tree-based Pipeline Optimization Tool)
H2O AutoML
AutoKeras
MLBox
Optuna
NNI (Neural Network Intelligence)
AutoGluon
Ray Tune
Determined AI
Hyperopt
Scikit-Optimize (skopt)
Spearmint
Bayesian Optimization
Nevergrad
BOHB (Bayesian Optimization and Hyperband)
GPflowOpt
Determined AI



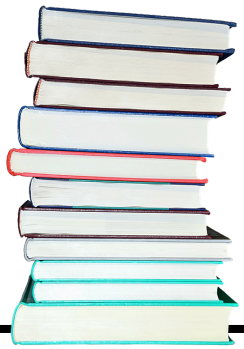
7. Evaluation

Steps

7.1 Evaluate Results
7.2 Review Process
7.3 Determine Next Steps

Python Libraries

Scikit-learn
Statsmodels
Yellowbrick
MLflow
TensorBoard
Weights & Biases
Optuna
Scikit-plot



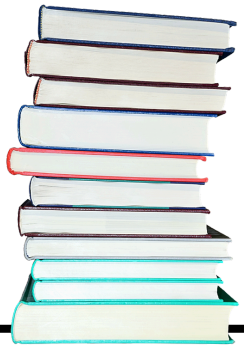
8. Deployment

Steps

8.1 Plan Deployment
8.2 Plan Monitoring and Maintenance
8.3 Produce Final Report
8.4 Review Project

Python Libraries

Streamlit
Flask
Django
FastAPI
Dash
Tornado
Bottle
Pyramid
Voilà
Anvil



9. AI Explainability

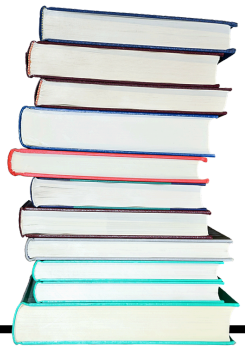
AI Explainability ensures that model decisions can be understood, questioned, and validated.

Steps

9.1 Understanding Explainability Needs
9.2 Select Explainability Techniques
9.3 Implement Explainability
9.4 Validate Explainability Outcomes
9.5 Monitor and Update Explainability
9.6 Document and Share Explainability Insights

Python Libraries

LIME (Local Interpretable Model-agnostic Explanations)
SHAP (SHapley Additive exPlanations)
Alibi
InterpretML
ELI5
What-If Tool
Skater
AIX360 (AI Explainability 360)
Fairlearn
PyTorch Captum
DALEX



10. AI Fairness

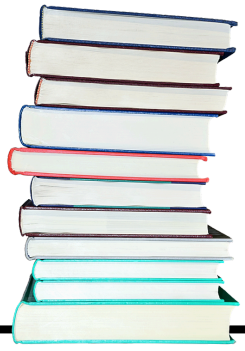
AI fairness is about developing ethical and trustworthy AI systems and preventing biases.

Steps

10.1 Understanding Fairness Requirements
10.2 Assess Data for Bias
10.3 Select Fairness Metrics and Techniques
10.4 Implement Fairness Measures
10.5 Validate Fairness Outcomes
10.6 Monitor and Update for Fairness
10.7 Document and Share Fairness Efforts

Python Libraries

Fairlearn
AI Fairness 360 (AIF360)
What-If Tool
EthicalML/XAI
Themis-ml
FairTest
Scikit-fairness
Fairness Comparison
Fairkit-learn
DEON



MARYAM MIRADI

YOUR CHALLENGE

DO YOU FIND IT CHALLENGING TO APPLY THEORETICAL AI AND DATA SCIENCE KNOWLEDGE TO REAL-WORLD SCENARIOS?

ARE YOU OVERWHELMED BY THE VAST AMOUNT OF AI RESOURCES, UNSURE WHERE TO START OR HOW TO PROGRESS EFFECTIVELY?

I HAVE THE RIGHT TRAINING FOR YOU.

AI Solutions Mastery bridges the gap between Theory and Practice by focusing on real-world applications.

AI Solutions Mastery

AI Practitioner Plus

You Learn

- End-to-End Data Science Implementation with Real-World Data (10-Steps)
- Advanced Feature Engineering (incl. Graph Analytics)
- Handling Imbalanced Data and Synthetic Data Generation (GANs)
- Supervised Learning (XGBoost and more)
- Deep Learning (TensorFlow Keras)
- Unsupervised Anomaly Detection (Isolation Forest, AutoEncoders)
- Unsupervised Clustering (DBSCAN, Hierarchical Clustering)
- AI Explainability and Fairness (SHAP, FairLearn)
- Model Deployment (Streamlit)

https://www.maryammiradi.com/training

MARYAM MIRADI

HANDS-ON TRAINING TO

Build a Real-World AI Product in 4 Weeks

From Problem to Product

20+ Years of My AI experience in One Training

VIEW TRAINING

www.maryammiradi.com

References

https://github.com/dslp/dslp/blob/main/steps.md
https://towardsdatascience.com/7-evaluation-metrics-for-clustering-algorithms-bdc537ff54d2
https://towardsdatascience.com/three-performance-evaluation-metrics-of-clustering-when-ground-truth-labels-are-not-available-ee08cb3ff4fb
https://medium.datadriveninvestor.com/evaluation-metrics-for-clustering-96dcdbea437d
https://towardsdatascience.com/a-comprehensive-beginners-guide-to-the-diverse-field-of-anomaly-detection-8c818d153995
GitHub: feature-engine/feature_engine
GitHub: iskandr/fancyimpute
GitHub: ResidentMario/missingno
GitHub: scikit-learn-contrib/category_encoders
GitHub: scikit-learn-contrib/imbalanced-learn
GitHub: yzhao062/pyod
GitHub: great-expectations/great_expectations
GitHub: pandas-profiling/pandas-profiling
GitHub: ironmussa/Optimus
GitHub: FeatureLabs/featuretools
GitHub: blue-yonder/tsfresh
GitHub: pola-rs/polars
GitHub: scikit-learn-contrib/category_encoders
GitHub: pycaret/pycaret
GitHub: scikit-learn-contrib/boruta_py
GitHub: scikit-learn-contrib/sklearn-pandas
GitHub: hi-primus/optimus
GitHub: cod3licious/autofeat
GitHub: tensorflow/tensorflow
GitHub: pytorch/pytorch
GitHub: dmlc/xgboost
GitHub: microsoft/LightGBM
GitHub: catboost/catboost
GitHub: h2oai/h2o-3
GitHub: keras-team/keras
GitHub: fastai/fastai
GitHub: optuna/optuna
GitHub: explosion/spaCy
GitHub: huggingface/transformers

References

GitHub: scikit-learn/scikit-learn
GitHub: statsmodels/statsmodels
GitHub: DistrictDataLabs/yellowbrick
GitHub: mlflow/mlflow
GitHub: Part of TensorFlow: tensorflow/tensorflow
GitHub: wandb/client
GitHub: optuna/optuna
GitHub: reiinakano/scikit-plot
GitHub: automl/auto-sklearn
GitHub: EpistasisLab/tpot
GitHub: h2oai/h2o-3
GitHub: keras-team/autokeras
GitHub: AxeldeRomblay/MLBox
GitHub: microsoft/nni
GitHub: awslabs/autogluon
GitHub: ray-project/ray
GitHub: determined-ai/determined
GitHub: hyperopt/hyperopt
GitHub: scikit-optimize/scikit-optimize
GitHub: ray-project/ray
GitHub: fmfn/BayesianOptimization
GitHub: facebookresearch/nevergrad
automl/HpBandSter
GitHub: GPflow/GPflowOpt.
GitHub: determined-ai/determined
GitHub: HIPS/Spearmint.
GitHub: streamlit/streamlit
GitHub: pallets/flask
GitHub: django/django
GitHub: tiangolo/fastapi
GitHub: plotly/dash
GitHub: tornadoweb/tornado
GitHub: bottlepy/bottle
GitHub: Pylons/pyramid
GitHub: voila-dashboards/voila
Web: Anvil

References

GitHub: oracle/Skater
GitHub: Trusted-AI/AIX360
GitHub: fairlearn/fairlearn
GitHub: pytorch/captum
GitHub: ModelOriented/DALEX
GitHub: marcotcr/lime
GitHub: slundberg/shap
GitHub: SeldonIO/alibi
GitHub: interpretml/interpret
GitHub: TeamHG-Memex/eli5
GitHub: fairlearn/fairlearn
GitHub: Trusted-AI/AIF360
GitHub: EthicalML/xai
GitHub: ayrna/themis-ml
GitHub: columbia/fairtest.
GitHub: scikit-learn-contrib
GitHub: algofairness/fairness-comparison
GitHub: fairkit-learn/fairkit-learn
GitHub: drivendataorg/deon